Advanced Statistical Means

Trimmed, Winsorized, Geometric, and Harmonic Analysis

1. Trimmed & Winsorized Mean

Used to reduce the influence of outliers or extreme values in a dataset.

Dataset: {2, 10, 12, 14, 15, 18, 100} (N=7)
Standard Mean: 24.4 (Heavily skewed by 100)

Trimmed Mean (20% Trim)

Removes a fixed percentage of the smallest and largest values.

- **Process:** Remove the top and bottom 20%. In this set, we remove 2 and 100.
- **Calculation:** (10 + 12 + 14 + 15 + 18) / 5
- **Result:** 13.8

Winsorized Mean (20% Winsorization)

Replaces extreme values with the nearest remaining values instead of deleting them.

- **Process:** Replace 2 with 10 and 100 with 18.
- **New Dataset:** {10, 10, 12, 14, 15, 18, 18}
- **Calculation:** (97) / 7
- **Result:** 13.86

2. Geometric Mean

The average of a set of products, primarily used for data involving growth rates, percentages, or ratios.

$$GM = \sqrt[n]{(x_1 \times x_2 \times ... \times x_n)}$$

Scenario: Investment growth over 3 years: 5%, 10%, and 20%.
Factors: 1.05, 1.10, 1.20

- **Calculation:** $\sqrt[3]{(1.05 \times 1.10 \times 1.20)} = \sqrt[3]{(1.386)}$
- **Result:** 1.1149 (Average growth of ~11.5%)

3. Harmonic Mean

The reciprocal of the arithmetic mean of the reciprocals. Used for rates and ratios, especially speed.

$$HM = n / \Sigma(1 / x_i)$$

Scenario: A car travels 10 miles at 40 mph and returns the same 10 miles at 60 mph.
Values: 40, 60 (n=2)

- **Calculation:** 2 / (1/40 + 1/60) = 2 / (0.025 + 0.0166) = 2 / 0.04166
- **Result:** 48.0 mph (The true average speed for the trip)